

Central Limit Theorem

Suppose we have a random variable X with an exponential distribution whose mean is $\mu = 3$.

$$\mu = 3$$

`data_exp =`

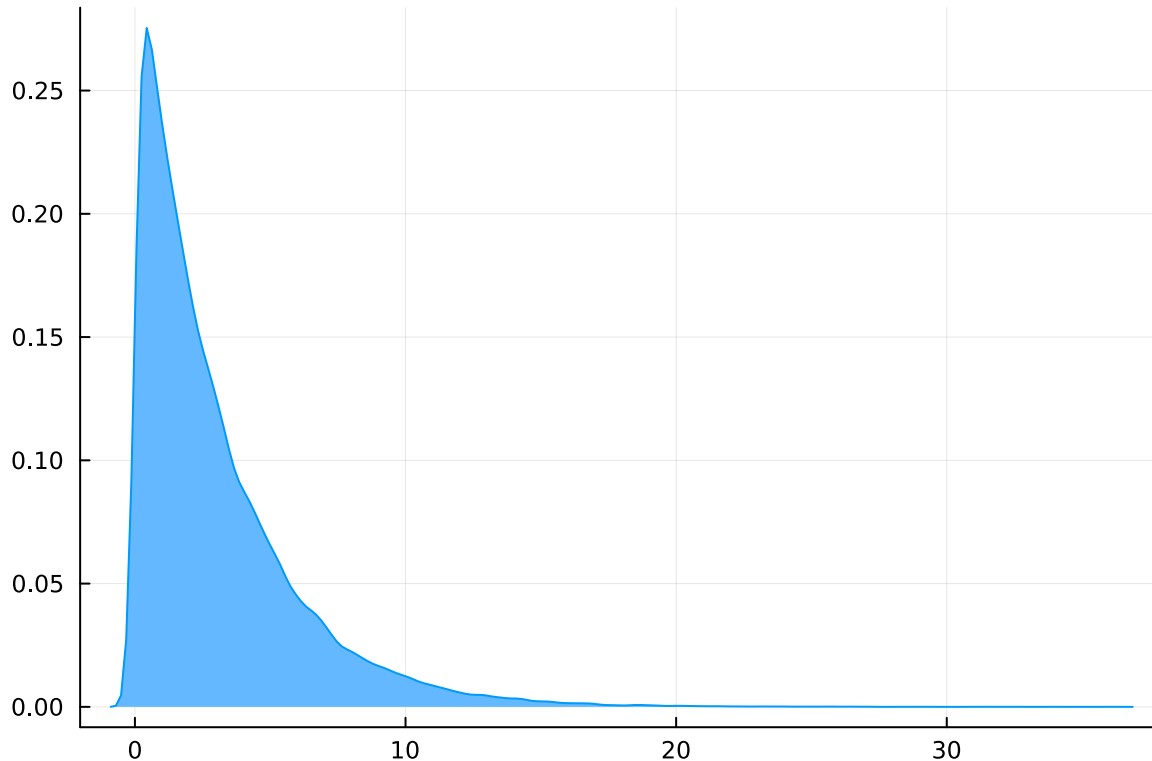
[27.0792, 9.22855, 5.5301, 10.5732, 6.35118, 4.70798, 3.2701, 0.898132, 4.66627, 0.774593, 0.6

$\bar{x}$ and s are the actual mean and standard deviations for the exponential data we generated above.

$$\bar{x} = 3.00802432112655$$

$$s = 2.9899593907737945$$

The histogram of our random variable X is shown below:



What is a sampling distribution?

A *sampling distribution* for a random variable X is the distribution of the values for a given sample statistic for all samples of a specified sample size n .

The given sample statistic that we will be interested in is the sample mean which leads to this mouthful:

A *sampling distribution of sample means* for a random variable X is the distribution of sample means for all samples of a specified sample size n . It is denoted by $\bar{X}$.

We generate a sampling distribution from our random variable X above using a sample size shown below and then plot that sampling distribution.

Sample size $n =$  10

Sample	Mean
[2.45, 0.25, 0.41, 4.39, 1.24, 1.66, 2.32, 0.62, 0.54, 0.81]	1.47
[1.07, 5.78, 1.48, 3.46, 2.76, 7.25, 4.99, 5.19, 0.53, 2.32]	3.48
[0.99, 0.18, 6.83, 2.66, 0.26, 6.21, 5.32, 4.22, 6.32, 5.59]	3.86
[1.61, 6.93, 6.93, 4.43, 1.03, 2.95, 0.49, 1.67, 1.42, 0.34]	2.78
[8.06, 2.07, 1.94, 4.0, 1.45, 2.48, 2.65, 3.02, 0.73, 0.32]	2.67
[2.39, 1.52, 0.7, 1.87, 0.17, 0.91, 0.65, 1.67, 1.33, 5.69]	1.69
[0.07, 1.04, 0.14, 5.38, 0.18, 3.89, 0.31, 7.1, 1.37, 1.19]	2.07
[3.61, 3.01, 1.54, 1.58, 0.17, 8.02, 3.09, 4.41, 1.14, 3.11]	2.97
[3.86, 7.09, 0.91, 1.94, 2.33, 3.42, 0.73, 0.49, 3.35, 1.51]	2.56
[0.03, 10.41, 2.25, 4.42, 2.22, 0.14, 1.7, 2.19, 0.36, 3.84]	2.76
:	:

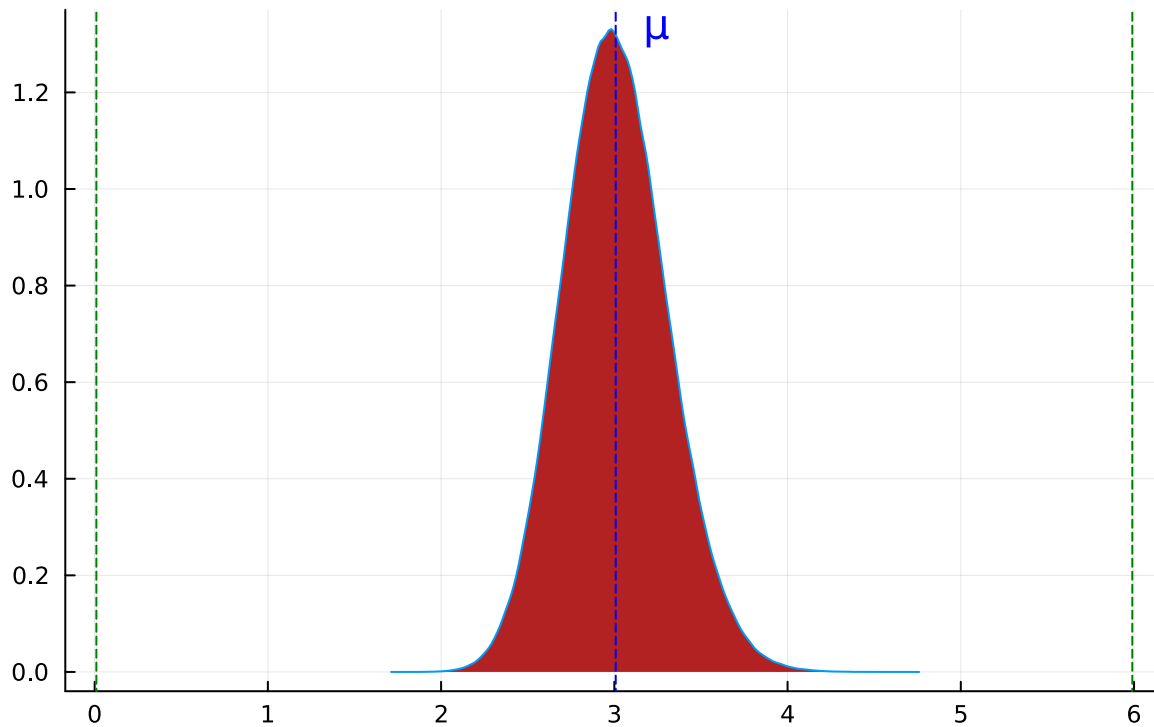
So the sampling distribution is

{1.47, 3.48, 3.86, 2.78, 2.67, 1.69, 2.07, 2.97, 2.56, 2.76, ...}

The sampling distribution we use below uses 1,000,000 samples since we can't actually find an infinite number of samples.

Sample size $n =$  99

sample size = 99



Watch the mode (highest peak) compared to the population mean (blue dashed line). [fps=2]

0:23 / 2:40

Now watch the standard deviation (standard error) as the sample size increases. [fps=10]

0:32 / 0:32

We can make the following observations about the sampling distribution:

- As the sample size increases the sampling distribution starts to approximate a normal distribution (usually $n = 30$ is enough) such that
- the sampling distributions mean is the same as original distributions mean
- and the sampling distributions standard deviation shrinks to zero. The standard deviation of the sampling distribution of sample means is called the standard error, but we won't use this terminology in this class.